

Dennis J Carroll

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Independent mechanistic interpretability researcher who traces model behavior to the circuit level before trusting it. A four-paper study (activation patching, logit-lens analysis, and steering interventions on GPT-2 Small through 1.5B, built on TransformerLens) locates the specific attention heads that override in-context evidence with training-frequency priors, and honestly reports where mechanistically-derived steering vectors fail to correct the bias. Approaches hallucination as a calibration problem: a model producing a frequency-prior answer over contradicting in-context evidence is a concrete, measurable instance of that failure mode. Currently completing an MDS at Illinois Institute of Technology with a Bayesian capstone ($R^2=0.481$, calibrated credible intervals). Foundation-first by habit: traces circuits and algorithms by hand before trusting library abstractions.

Core Competencies

- **Mechanistic Interpretability:** Activation patching, logit-lens analysis, activation steering (difference-of-means vectors), circuit-level analysis, TransformerLens
- **Statistical Analysis & Probabilistic Modeling:** Bayesian inference, hierarchical modeling, uncertainty quantification, posterior predictive checks, calibration analysis
- **Python & ML:** PyTorch, Scikit-learn, NumPy, Pandas; model development from scratch, hyperparameter tuning
- **Research Practice:** Hypothesis-driven experimentation, scale-sweep methodology, honest reporting of negative and limited results, technical writing, publishing research code openly on GitHub
- **Tools:** Git/GitHub, Jupyter, Linux, SQL, Claude Code (AI-assisted development)

Independent Research

- **GPT-2 (124M–1.5B), TransformerLens**
The Frequency Prior Trilogy – Mechanistic Interpretability 2025–2026
 - Located the circuit (heads L9H8, L8H11, L10H0) responsible for GPT-2 Small overriding in-context evidence with training-frequency priors; in-context accuracy degrades from 79% at $n=1$ examples to 55% at $n=5$ as the prior dominates
 - Tested activation-steering interventions to correct the bias: a mechanistically-derived difference-of-means vector corrected roughly one-fifth of cases at safe doses, while a black-box learned vector succeeded where the interpretable one failed, a limitation reported honestly rather than hidden
 - Extended the analysis across the full GPT-2 family and multiple semantic domains (languages, currencies, demonyms); found the bias is not one direction but a structured subspace (energy 0.45), with frequency bias clearing by 345M parameters while frequency absence itself persists to 1.5B
 - Built Attn Flow, a browser-based attention-flow visualizer using particle simulation to trace probability mass through the residual stream; dennisjcarroll.com/research
- **Python, Probabilistic Modeling**
Bayesian Analysis of Seattle Airbnb Market 2025
 - Built hierarchical Bayesian models with varying intercepts and slopes by neighborhood; achieved $R^2=0.481$ with calibrated credible intervals and posterior predictive checks
 - Quantified epistemic uncertainty in pricing predictions, producing estimates that communicate confidence levels rather than false precision

Open-Source Engineering

- **Python 3.10+, zero runtime dependencies**
GLASSPORT – MCP Security Auditing Framework 2026
 - Wire-level observability and enforcement for Model Context Protocol servers: passive stdio tap, behavioral detectors (fabricated tool calls, context violations, data exfiltration), and static source audit with SARIF 2.1.0 export to the GitHub Security tab
 - 317 passing tests; PII scanning hardened against ReDoS and obfuscation, with fault-isolated detectors; github.com/Dennis-J-Carroll/glassport
- **JavaScript, TensorFlow.js, React, Gatsby**
Interactive Browser Applications 2021–Present
 - Built 26 standalone browser apps — neural-network visualizers, Bayesian analytics dashboards, real-time ML training environments — all static, no install; dennisjcarroll.com

Education

- **Illinois Institute of Technology** Chicago, IL
Master of Data Science (in progress, part-time) May 2024–present
Coursework: Machine Learning, Bayesian Inference, NLP, Probabilistic Modeling, Distributed Computing, Data Mining, Predictive Analytics.
- **SUNY Oneonta** Oneonta, NY
Bachelor of Fine Arts in Communication Studies 2013–2017

Professional Experience

- **60th & Fifth Corp (32BJ SEIU)** New York, NY
Doorman Mar 2020–Present
 - Manage complex resident database with 40+ families while completing graduate coursework and independent research part-time
- **California Closets** Hawthorne, NY
Carpenter Assistant Mar 2019–Mar 2020
- **Lowe's Home Improvement** Orangeburg, NY
Paint Department Associate May 2018–Mar 2020

Certifications

- **Applied AI and Data Science** – Brown University School of Professional Studies (2024)
- **Social & Behavioral Responsible Conduct of Research** – CITI Program (2023, valid through 2026)

References

- Available upon request.